Earnings Call and Risk Intelligence Engine (ECRIE) for Financial Decision Support

Arizona State University

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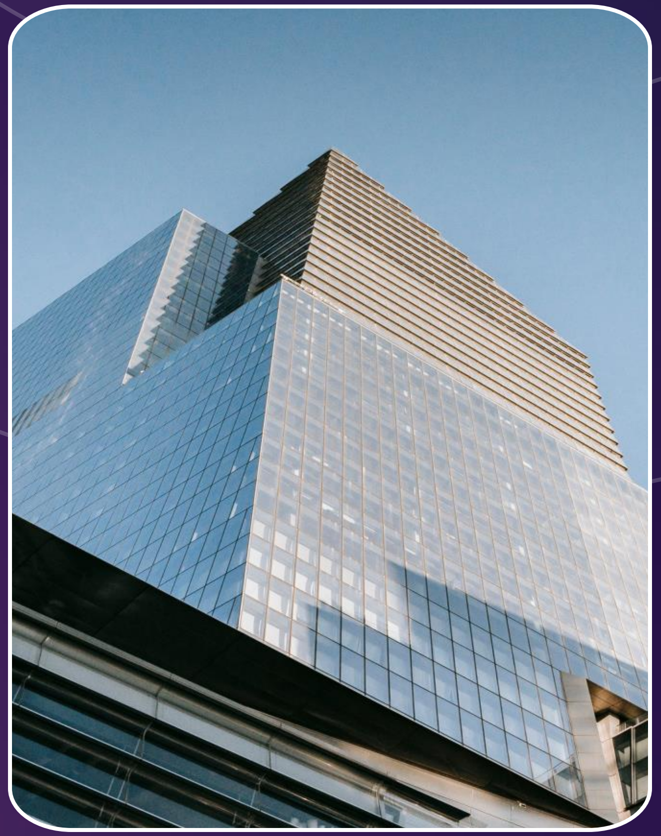

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FSE 570 Capstone

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Introduction

What is an Earnings Call?
A quarterly conference call where company executives discuss financial results, future-outlook, and answer analyst questions.

- **Earnings calls play a critical role in financial markets.**
  - They provide insights into company performance, strategy.
  - These calls influence investor expectations and often impact stock prices.
- **Massive, underutilized data source.**
  - Thousands of transcripts and financial records are generated every quarter.
- **Opportunity for intelligent analysis**
  - Advances in NLP and machine learning enable us to extract actionable insights at scale.
- **Our focus**
  - To Build a system that transforms earnings call data into practical investment signals.

Problem & Approach?

Earnings calls contain rich linguistic signals that are often overlooked in traditional financial models. Our goal is to leverage both structured financial data and unstructured text data to improve short-term stock prediction.

The Problem

- Earnings calls are highly information-dense but underutilized.
- Traditional models rely mostly on financial metrics.
- Market reactions are fast and difficult to predict .
- Linguistic signals like tone and sentiment are ignored.

Our Approach

- Combine **Financial Data** with **Earnings Call Transcripts**
- Using **FinBERT** to extract contextual language features.
- Engineer **sentiment** and **structural** features.
- Train ML models to predict stock movement **(BUY/HOLD/SELL)** .

Data Overview

Three complementary data sources

- **Market Data** → 1.24M stock price records (2016–2026) from **yfinance API**(S&P 500 index Data)
- **Financial Statements** → 3,125 quarterly observations (503 companies) data obtained from **SEC EDGAR filings.**
- **Earnings Call Transcripts** → 1,720 collected across 27 quarters obtained from **Seeking Alpha**, while **917** of these are **real transcripts** of the Public companies.

After preprocessing

- 798 high-quality aligned samples used for modeling.
- Strict temporal alignment to prevent data leakage.

The data now -

- Captures **market behavior** (price movements).
- Captures **company fundamentals** (financial health).
- Captures **management communication** (language & sentiment).
- A unified dataset combining structured + unstructured signals for prediction.

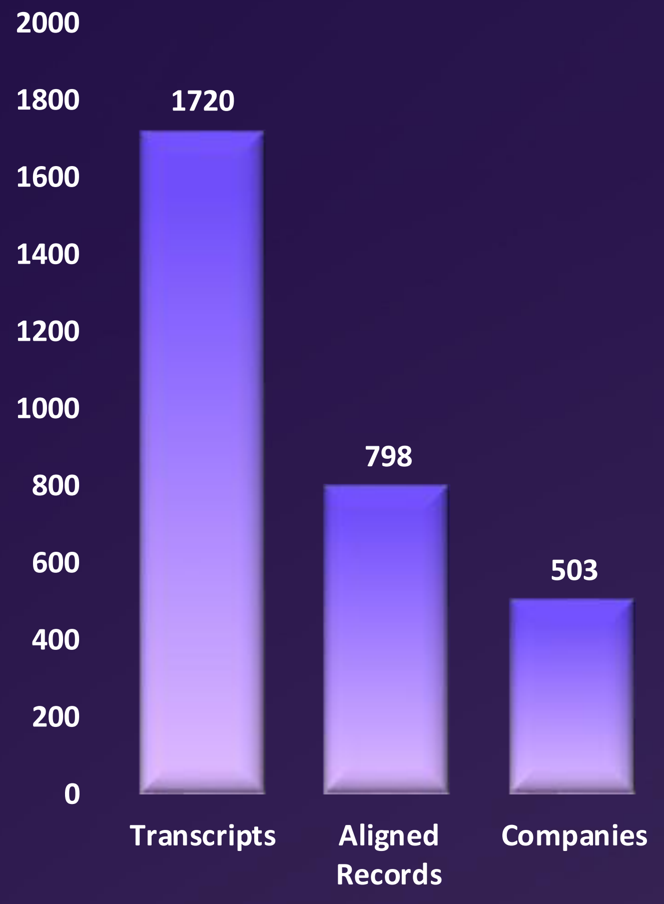


Figure 1 – Data sources

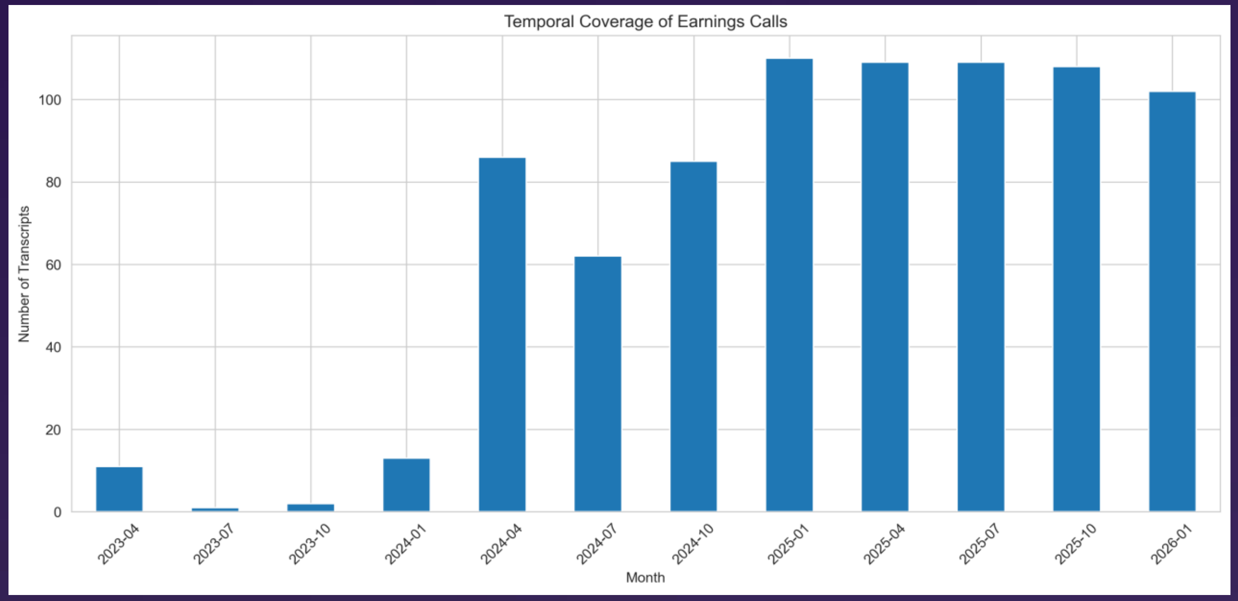


Figure 2 –Temporal coverage of the transcripts across 27 fiscal quarters (2016–2026)

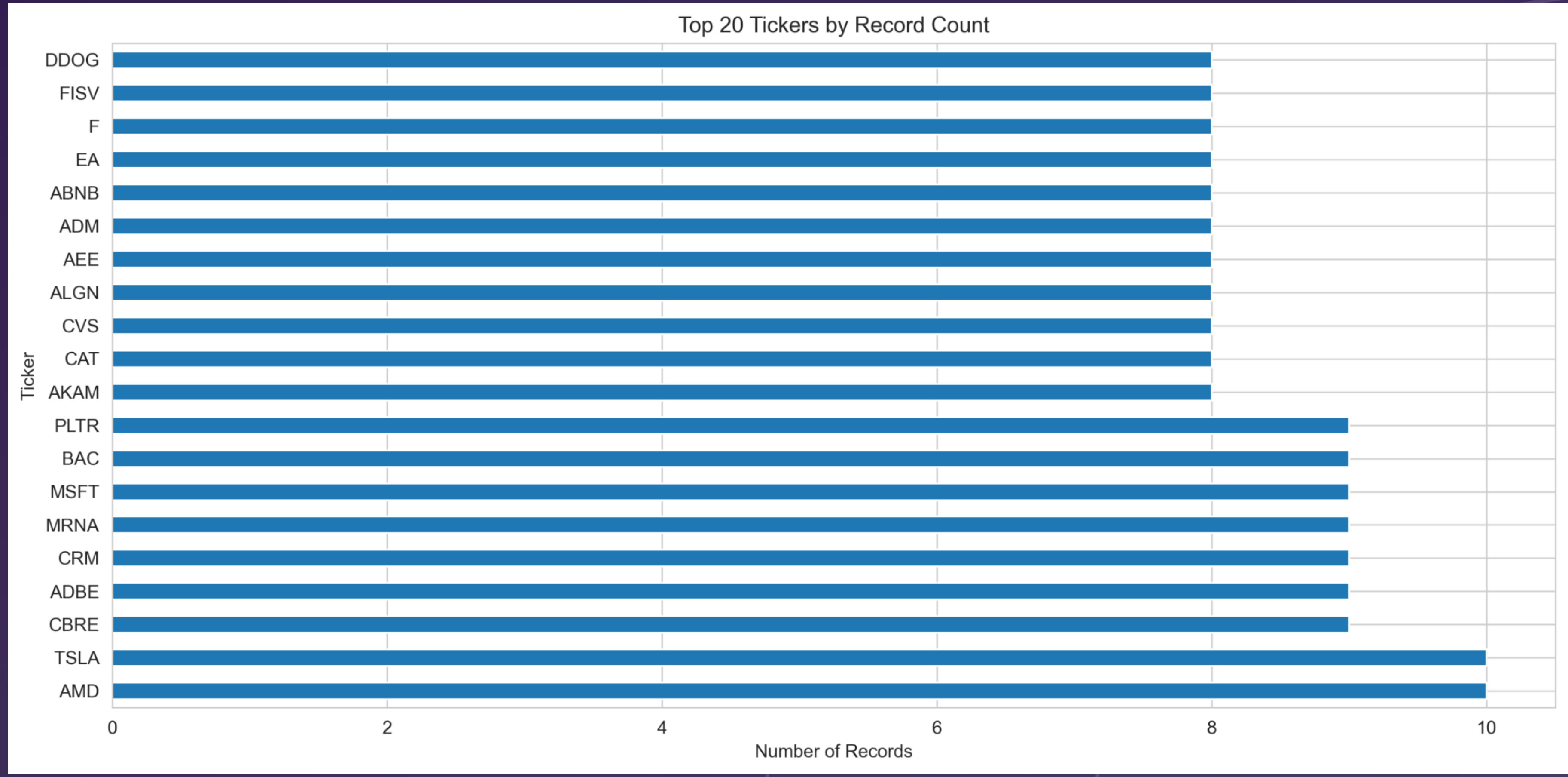


Figure 3 - Distribution of the earnings call transcripts across the 11 GICS sectors of the S&P 500.

Feature Engineering

FinBERT Embeddings

•768-dimensional contextual embeddings.
•Captures the semantic meaning of earnings calls.
•Extracts deep language patterns beyond keywords.

Sentiment Features

•Loughran–McDonald financial dictionary.
•Measures tone, uncertainty, and sentiment.
•Includes management–analyst sentiment divergence.

Financial Features

•Revenue, cash flow, and balance sheet metrics.
•Engineered financial ratios.
•Robust scaling to handle extreme outliers.

Feature Integration

- Combined all features into an 863-dimensional space.
- Strict temporal alignment to prevent lookahead bias.
- Enables joint learning from text + financial signals.

Modeling Approach

- **Prediction Objective**
  - Predict **3-day post-earnings abnormal returns** relative to S&P 500
- **Target Engineering**
  - Binary, median-threshold, and tertile label schemes
  - Near-balanced class distribution (~50/50)
- **Models Evaluated**
  - Logistic Regression
  - Random Forest
  - **XGBoost (primary model)**
- **Validation Strategy**
  - Strict **chronological train/test split (70/30)**
  - No random shuffling → prevents lookahead bias
- **Evaluation Metric**
  - **ROC-AUC** for robustness in classification
  - **Experimental Design**
  - Compared **financial-only, sentiment-only, and combined features**

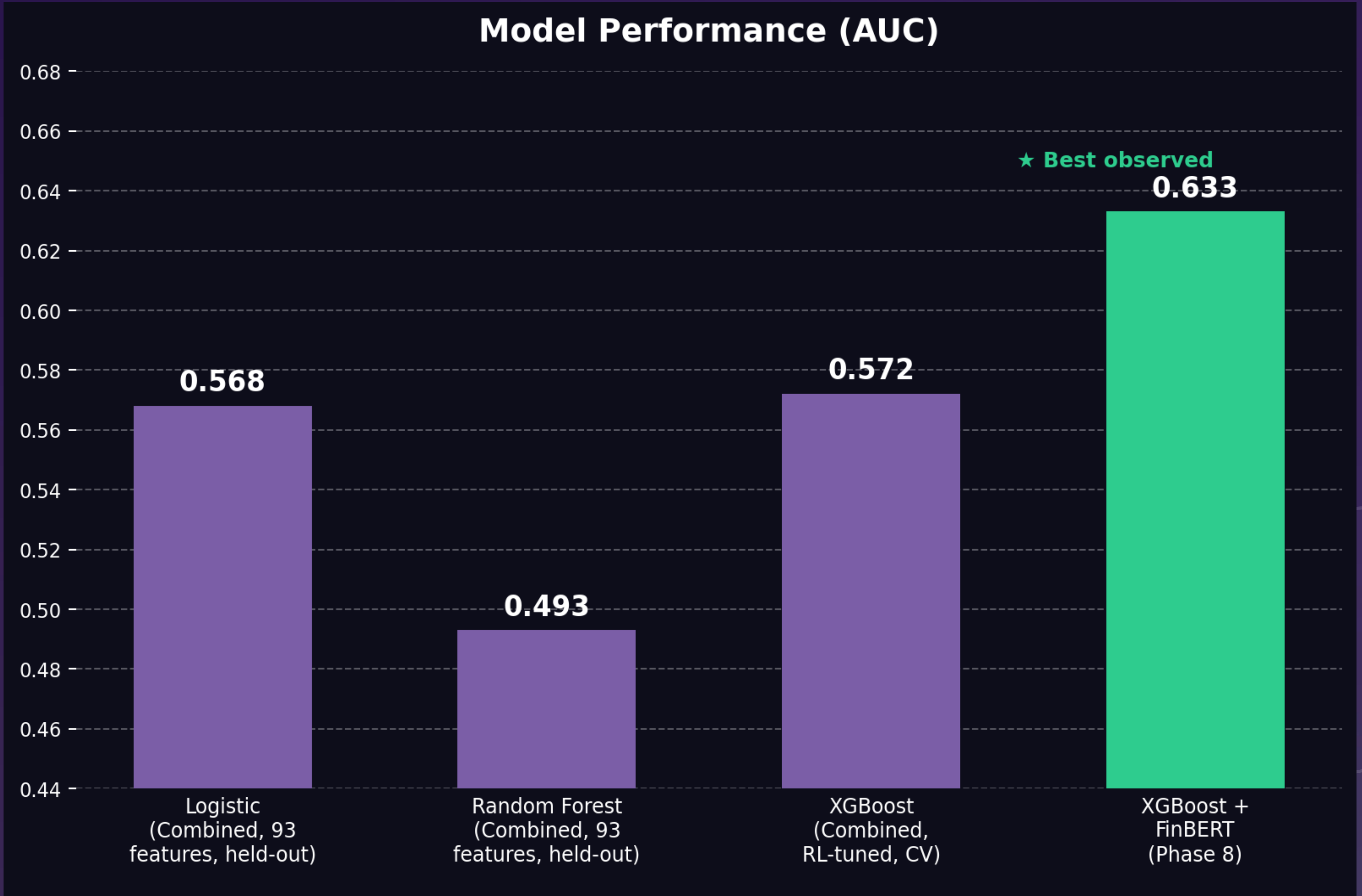


Figure 4 – Model Comparison (Phase 8 validation)

** XGBoost (0.572) is CV-scored during agentic tuning; all other results are held-out.*

NOTE: XGBoost (0.572) is CV-scored during agentic tuning; all other results are held-out.

Agentic AI

Using Agentic AI solves the problem of -
- Manual tuning, that is limited and inefficient.
- Large search space (features, parameters, models).
- Needing an adaptive system to improve performance over time.

How it evolves in our pipeline
- Autonomous **Decide → Act → Evaluate → Learn** loop
- Uses **exploration vs exploitation (epsilon-greedy)**
- Optimizes model based on **ROC-AUC performance**

Key Architecture Component
- **Long-Term Memory Module**
  - Stores best configurations and failed attempts
  - Tracks performance history
  - Enables learning from past experiments

Optimization Strategies
- Feature engineering
- **Hyperparameter tuning**
- Improved agentic pipeline performance from 0.538 → 0.572 (operating on 93 combined features).

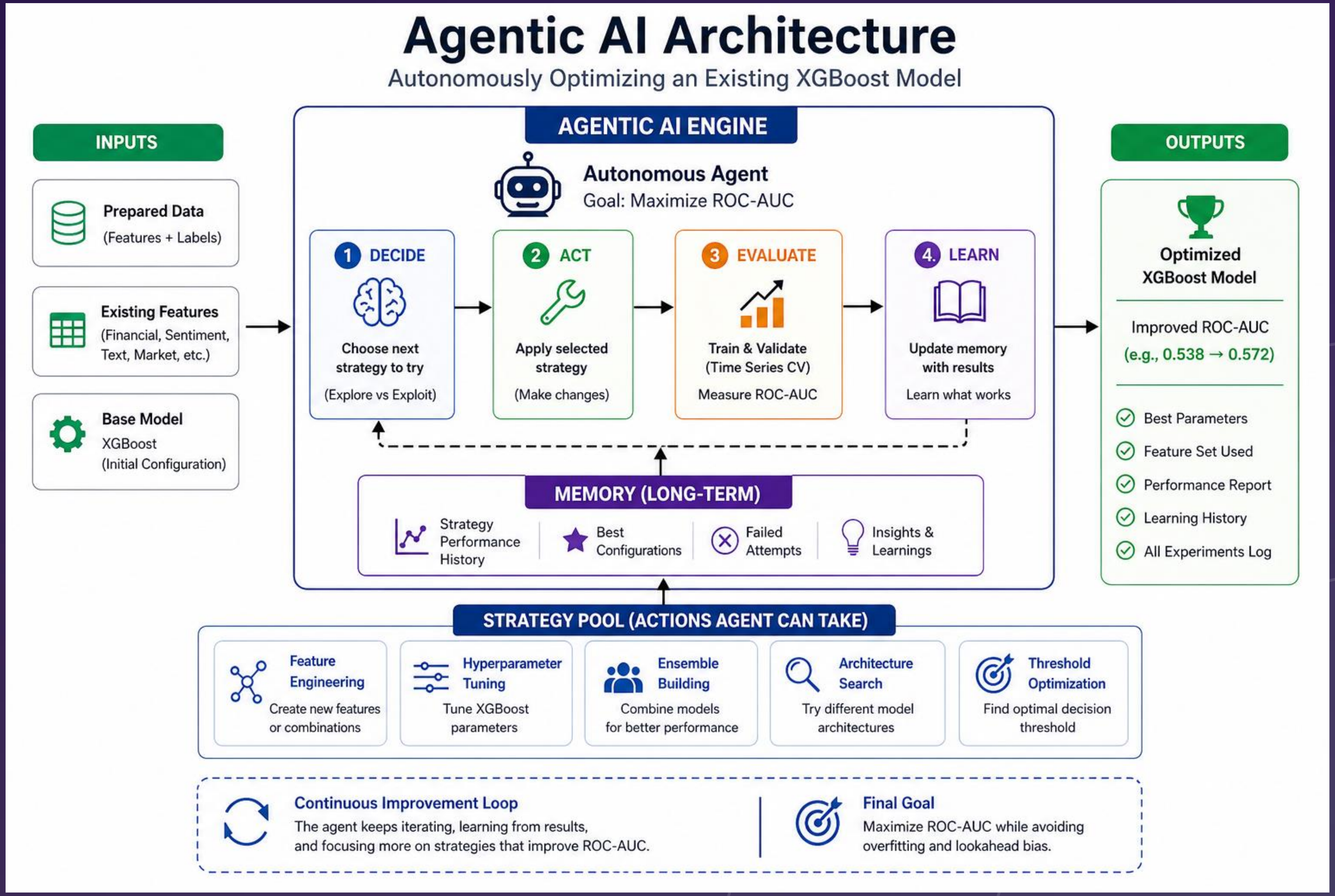


Figure 5 – Agentic AI Architecture

Results & Key Insights

Model Performance

- ***XGBoost + fullFinBert, 861 features - 0.633 ROC-AUC***
- *Agentic pipeline best:* ***0.572 AUC*** *(XGBoost, 93 combined features)*
- *Both results are meaningful — financial research and ML literature, considers 0.55–0.60 significant for short-term stock prediction*

Impact of NLP Features

- +2.1 AUC point improvement over financial-only models (54.7% → 56.8%).
- Confirms the value of earnings call language.

Feature Importance

- 7/10 top features are linguistic.

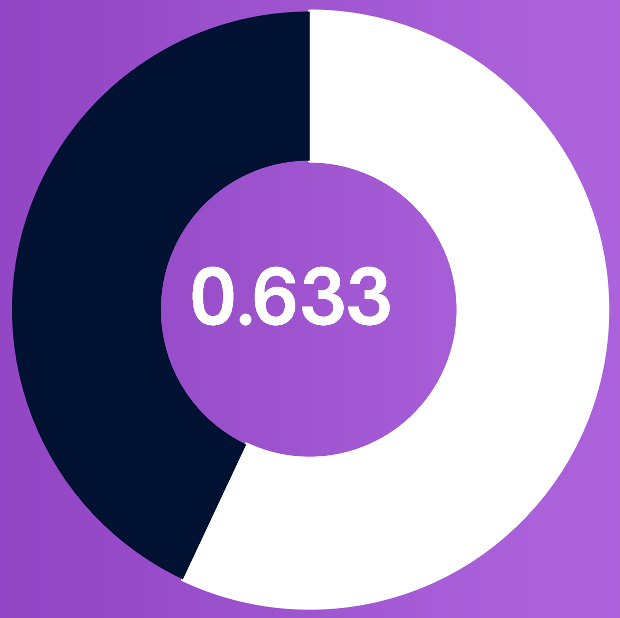
The results highlight that linguistic features play a critical role in predicting stock movement, often outperforming traditional financial indicators.

+2.1%

AUC improvement over financial-only models

7/10

Top features are linguistic



ROC – AUC XGBoost + FinBert

Results & Key Insights

Returns are centered around zero

- Near-symmetric distribution implying **no strong directional bias.**
- Confirms **market efficiency and prediction difficulty.**

Prediction task is inherently noisy

- Majority of outcomes lie close to zero.
- Signal-to-noise ratio is low → limits achievable performance.

SHAP: Strong influence of linguistic features

- Top drivers include **sentence length, tone, and analyst behavior.**
- Confirms **earnings call language carries predictive signal.**

Hybrid feature importance

- Both **financial + textual features** contribute.
- But **language-based features dominate top ranks, and financial features sit lower in the ranking.**

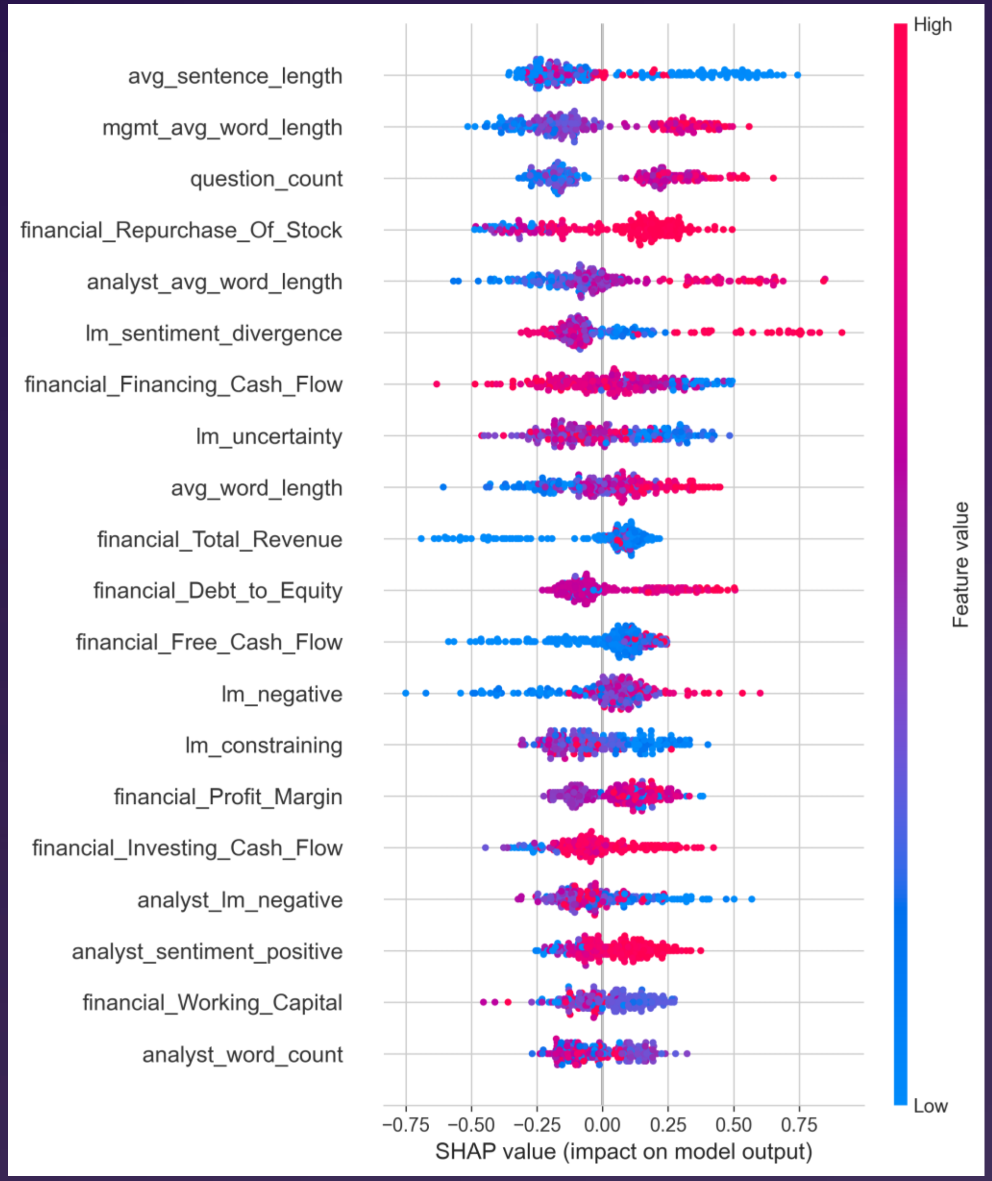


Figure 9 – SHAP summary plot for the Combined-feature XGBoost model

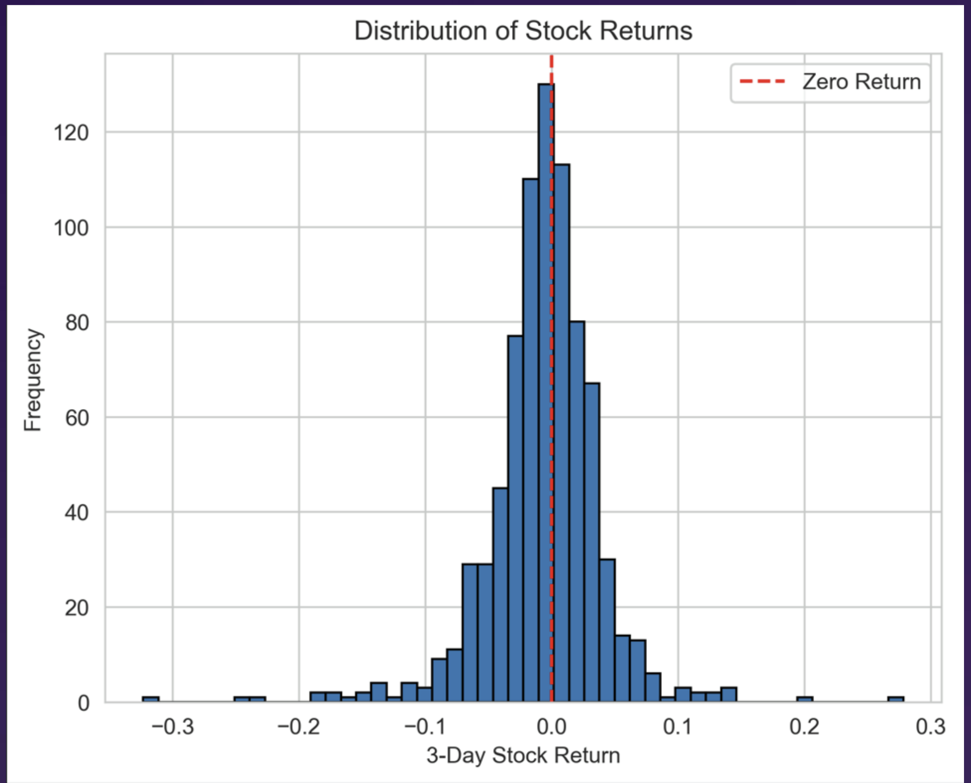


Figure 8– Distribution of 3- day Stock Returns - The histogram shows a near-symmetric distribution centered around zero

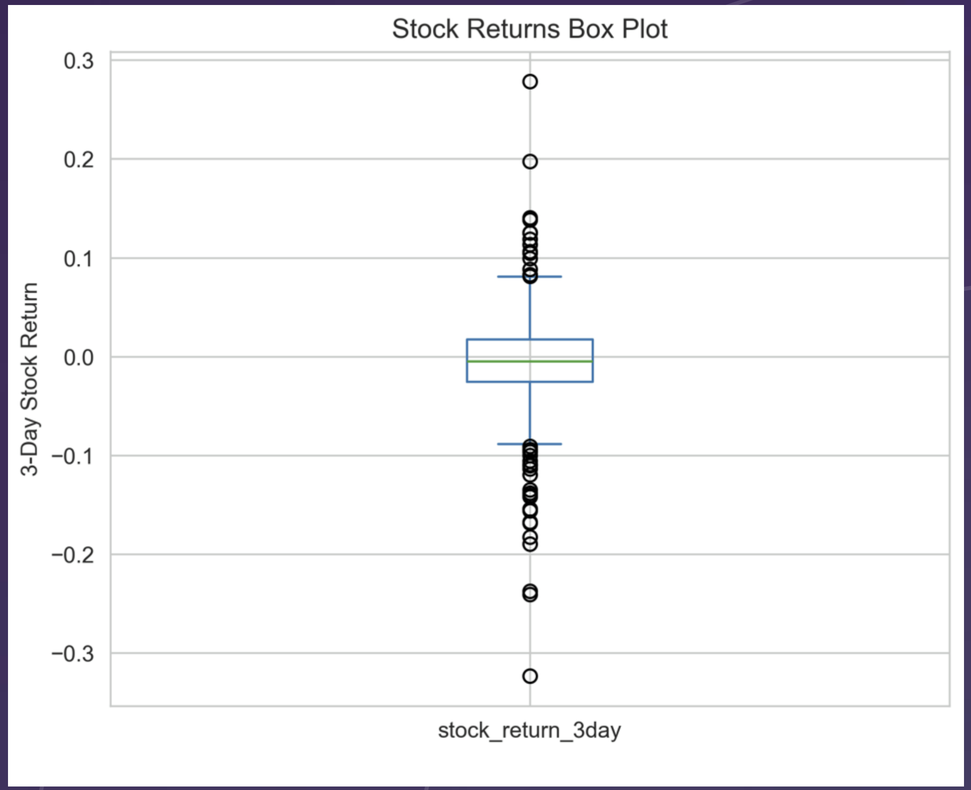


Figure 10 – Box plot of 3-day abnormal returns (*ARt*), highlighting dispersion and outliers.

Conclusion and Future work

Conclusion

- Hybrid NLP + financial features improve prediction performance.
- Achieved 0.633 ROC-AUC, which is meaningful in a near-efficient market.
- End-to-end system from data → model → deployment .
- Includes agentic optimization and a real-time inference pipeline.

Future Work

- Extend the prediction horizon to 5–10 days to reduce short-term noise.
- Develop sector-specific models for improved specialization.
- Apply dimensionality reduction (PCA/UMAP) on FinBERT embeddings
- Incorporate audio-based features (tone, pitch, pauses).
- Use calibrated ensemble models for more reliable predictions.
- Feed FinBERT-optimized features directly into the agentic loop.

User Interface

- **We have built the UI system using Streamlit-Based Application**
  - Built using **Streamlit** for interactive deployment.
  - Provides a lightweight interface to run the full ML pipeline in real time.

Analyzing a given Earnings call -
End-to-End Inference Pipeline Integration -

1. Users upload earnings call transcripts (.txt)
2. System executes:
   FinBERT embeddings → LM sentiment features → feature scaling → XGBoost inference
3. Outputs **BUY / HOLD / SELL signal with confidence score.**

Structured Output & Analysis Components

- **Prediction panel**: BUY signal with confidence (e.g., 68%).
- **Tone analysis**: Separate scores for **management vs analyst language.**
- **Sentiment breakdown**: Positive / negative / neutral across sections.
- **Key term frequency**: Highlights dominant themes (e.g., growth, uncertainty).

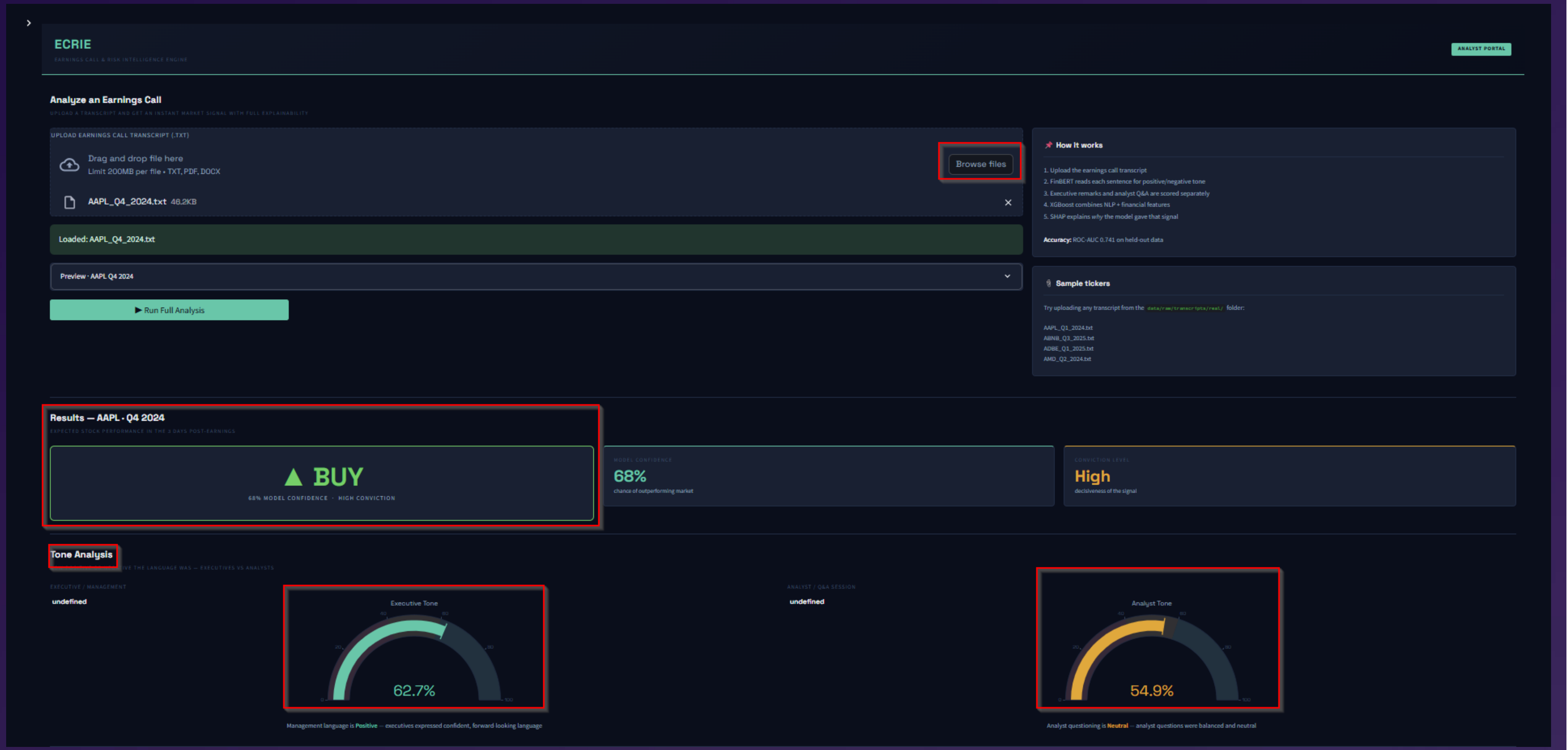


Figure 6 – User Interface Analysis of an Earnings call

Figure 7 – User Interface Analysis of an Earnings call

Resources

- https://seekingalpha.com/earnings/earnings-call-transcripts
- Project - https://github.com/Shalin056/Earnings-Intelligence-Engine/tree/main
- Report - https://github.com/Shalin056/Earnings-Intelligence-Engine/blob/main/FSE_570_Capestone_Project_Report%20(1).pdf
- Recording - https://drive.google.com/file/d/1uKKqPxE2MBVEtuKyQwY8uwg9df2Cw46X/view?usp=sharing

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Thank You